Bhuvan Sachdeva

https://bhuvan-21.github.io/ @ sachdeva.bhuvan21@gmail.com github.com/Bhuvan-21 Google Scholar

Education

May 2023 | **Delhi Technological University (DTU)** — **Delhi, India**
Aug 2019 | B.Tech - Computer Science and Engineering, **GPA: 9.22/10**

Research Experience

Present / **July 2026** | **Stellon Labs** — **Remote**
Member of Technical Staff
Building multimodal speech models to advance end-to-end audio understanding.

June 2026 / **Aug 2023** | **Sankara Eye Hospital | (Visiting Researcher) Microsoft Research** — **Bangalore, India**
Research Fellow | Advisors: Dr. Mohit Jain, Prof. Vineeth N B, Dr. Kaushik Murali
Worked on task transfer in vision–language models for perception tasks, complication detection in cataract surgery, and healthcare chatbots for patient support.

July 2022 / **May 2022** | **Prime Video, Amazon** — **Bangalore, India**
Applied Scientist Intern | Advisor: Prof. Anil Nelakanti
Improved machine translation for **Prime Video** system using Automatic Post Editing.

Mar 2023 / **June 2021** | **Kroop AI** — **Remote**
Research Intern | Advisor: Prof. Abhinav Dhall
Worked on projects involving face swapping, voice cloning, generating lip movements from audio and deep-fake detection using facial landmarks.

Select Publications

S=Under Review, C=Conference, W=Workshop, P=Poster/Demo, J=Journal

[C.3] **Understanding Task Transfer in Vision-Language Models** [🔗]
Bhuvan Sachdeva*, Karan Uppal*, Abhinav Java*, Vineeth N. B
IEEE/CVF Conference on Computer Vision and Pattern Recognition **(Oral)** | *UniReps @ Neurips'25* [**CVPR'26**]

[C.2] **Swift Sampling: Selecting Temporal Surprises via Taylor Series** [🔗]
Dahye Kim, Bhuvan Sachdeva*, Karan Uppal*, Naman Gupta*, Vineeth N. B, Deepti Ghadiyaram
Conference on Neural Information Processing Systems [**Neurips'26**)]

[W.1] **BAGEL-RL: Low-Rank Reinforcement Learning for Interleaved Visual Reasoning** [🔗]
Bhuvan Sachdeva, Vineeth N. B
Embodied Spatial Reasoning Workshop @ Neurips'26 [**ESR@Neurips'26**]

[C.1] **Phase-Informed Tool Segmentation for Manual Small-Incision Cataract Surgery** [🔗]
Bhuvan Sachdeva*, Naren Akash*, ..., Kaushik Murali, Mohit Jain
International Conference on Medical Image Computing and Computer Assisted Intervention [**MICCAI'25**]

[J.1] **CataractBot: An LLM-Powered Expert-in-the-Loop Chatbot for Cataract Patients** [🔗]
Pragnya Ramjee*, Bhuvan Sachdeva*, ..., Kaushik Murali, Mohit Jain
ACM Journal on Interactive, Mobile, Wearable and Ubiquitous Techonologies [**ACM IMWUT'25**]

Select Research Projects

Task Transfer in VLMs

- Finetuned Qwen-2.5VL models (3B, 7B and 32B) on 13 perception tasks to analyze cross-task performance transfer.
- Demonstrated asymmetric transfer patterns and transfer guided finetuning in low-data regime; accepted at **CVPR'26**.

BYOeB: Build Your Own Expert Bot

- Architected an open-source, expert-in-the-loop RAG framework supporting multilingual Indic queries and query rewriting via Microsoft at **BYOeB**.
- Deployed at scale as **CataractBot** (1k+ patients) and **ASHABot** (50k+ health workers); accepted at **IMWUT'25** and **CHI'25**.

Achievements and Awards

Oral at CVPR 2026: Paper accepted for oral presentation (top-1% of papers) at CVPR, 2026 — 2026

Received **MICCAI 2025 Young Scientist** Registration Grant — 2025

Candidate Master on Codeforces (1900+ rating): _bhuvan21 — 2021